Gas Problems: 10-Problem Set

Work on scratch paper or in the spaces. Answers and full worked solutions are in *gas_10_key.pdf*. ★ = exam-level problem. Use R = 0.08206 L·atm/(mol·K) unless told otherwise.

THE 5-STEP GAS ROUTINE (do this for EVERY gas problem)

1. **Make a table** of P, V, n, T. Write each given value with its unit and put "?" for the one you want. Two states (before/after)? Make two columns.
2. **Convert**: T → K (+273.15) · P → atm (÷760 for mmHg/Torr, ÷101.325 for kPa) · V → L (mL ÷ 1000) · grams → mol (÷$\mathcal{M}$).
3. **Pick the equation**:

Situation	Use
One state, find one of P, V, n, T	PV = nRT
Same gas, conditions change (before → after)	$P_1V_1/(n_1T_1) = P_2V_2/(n_2T_2)$; cross out what stays constant
Grams + volume given, find $\mathcal{M}$ or density	$\mathcal{M}$ = mRT/(PV) · d = P$\mathcal{M}$/(RT)
Mixture of gases	$P_{tot} = \Sigma P_i$ · $P_i = \chi_i \cdot P_{tot}$ · $P_i = n_iRT/V$
Collected over water	$P_{gas} = P_{total} - P_{H_2O}$
Reaction involving a gas	gas ⇄ mol via PV = nRT, then mole ratio. Same T and P? volume ratio = mole ratio
Rates / effusion	$rate_1/rate_2 = \sqrt{(\mathcal{M}_2/\mathcal{M}_1)}$ · $time_1/time_2 = \sqrt{(\mathcal{M}_1/\mathcal{M}_2)}$
Speeds / kinetic energy	$u_{rms} = \sqrt{(3RT/\mathcal{M})}$, R = 8.314, **$\mathcal{M}$ in kg/mol** · KE_avg = 3/2 RT (per mol)

4. **Predict first**: does the answer go up or down? (e.g. hotter → bigger V; more pressure → smaller V)
5. **Solve, then check** units, sig figs, and that the direction matches your prediction.

Problems

1. ★ (Combined) A weather balloon holds 5.00 L of He at 755 mm-Hg and 25 °C at launch. It rises to an altitude where P = 380. mm-Hg and T = −15 °C. What is its volume there? Explain why it expanded even though it got colder.

2. 4.50 g of CO_2 is placed in a 2.00-L flask at body temperature (37 °C). What is the pressure in atm?

3. ★ A hydrocarbon contains 85.6% C and 14.4% H by mass. A 0.802-g sample of its vapor occupies 354 mL at 100. °C and 752 mm-Hg. Determine the molar mass and the molecular formula.

4. ★ A 5.00-L flask at 25 °C contains 2.00 g He and 8.00 g O_2. Find the partial pressure of each gas, the total pressure, and the mole fraction of He.

5. ★ Heating $KClO_3$ gives O_2: $2\ KClO_3(s) \rightarrow 2\ KCl(s) + 3\ O_2(g)$. The O_2 is collected over water: 0.512 L at 22 °C and a total pressure of 758 mm-Hg (vapor pressure of water at 22 °C = 19.8 mm-Hg). What mass of $KClO_3$ (122.55 g/mol) decomposed?

6. ★ Car airbags inflate by $2\ NaN_3(s) \rightarrow 2\ Na(s) + 3\ N_2(g)$. What mass of NaN_3 (65.01 g/mol) is needed to fill a 65.0-L airbag at 1.10 atm and 25 °C?

7. 15.0 L H_2 and 6.00 L N_2 react at the same T and P: $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$. Which is limiting? What volume of NH_3 forms, and what volume of which gas is left over? (No ideal gas law needed!)

8. ★ 2.00 g of Al (26.982 g/mol) is reacted with 1.50 L of Cl_2 gas at 25 °C and 1.00 atm: $2\ Al(s) + 3\ Cl_2(g) \rightarrow 2\ AlCl_3(s)$. Determine the limiting reactant and the mass of $AlCl_3$ (133.33 g/mol) formed.

9. (a) How many times faster does H_2 effuse than O_2? (b) An unknown gas effuses 0.354 times as fast as He. What is its molar mass, and what could it be? (c) 1.00 L of N_2 effuses through a pinhole in 105 s; the same volume of an unknown gas takes 191 s. Find the unknown's molar mass.

10. (a) Calculate u_rms for He and for Xe (131.29 g/mol) at 25 °C. (b) At what temperature would O_2 molecules have the same u_rms as He atoms at 25 °C?